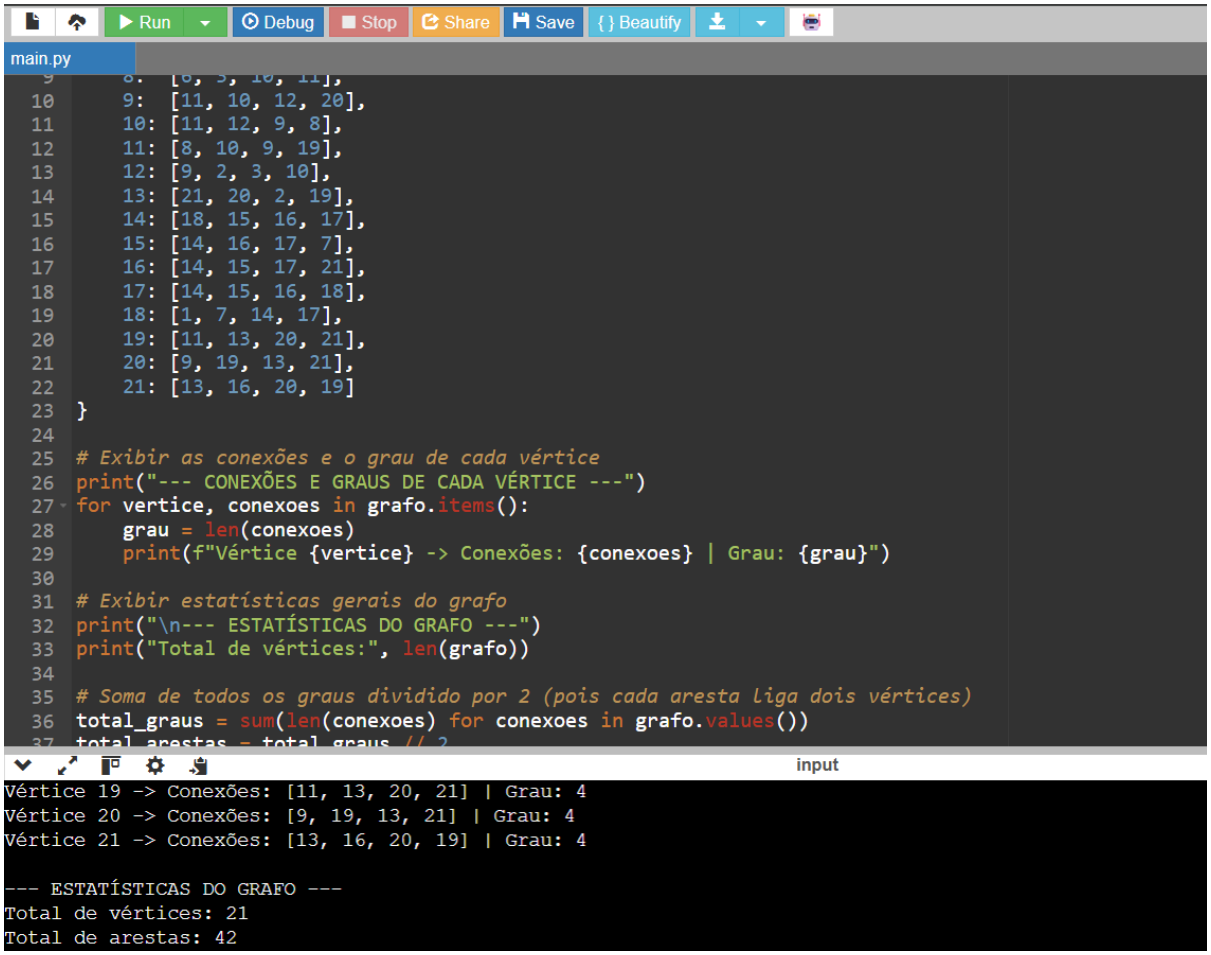


grafo = {

1: [7, 18, 2, 5],

2: [1, 4, 12, 13],

3: [6, 12, 4, 8],

4: [5, 6, 3, 2],

5: [4, 7, 1, 6],

6: [4, 5, 3, 8],

7: [5, 18, 1, 15],

8: [6, 3, 10, 11],

9: [11, 10, 12, 20],

10: [11, 12, 9, 8],

11: [8, 10, 9, 19],

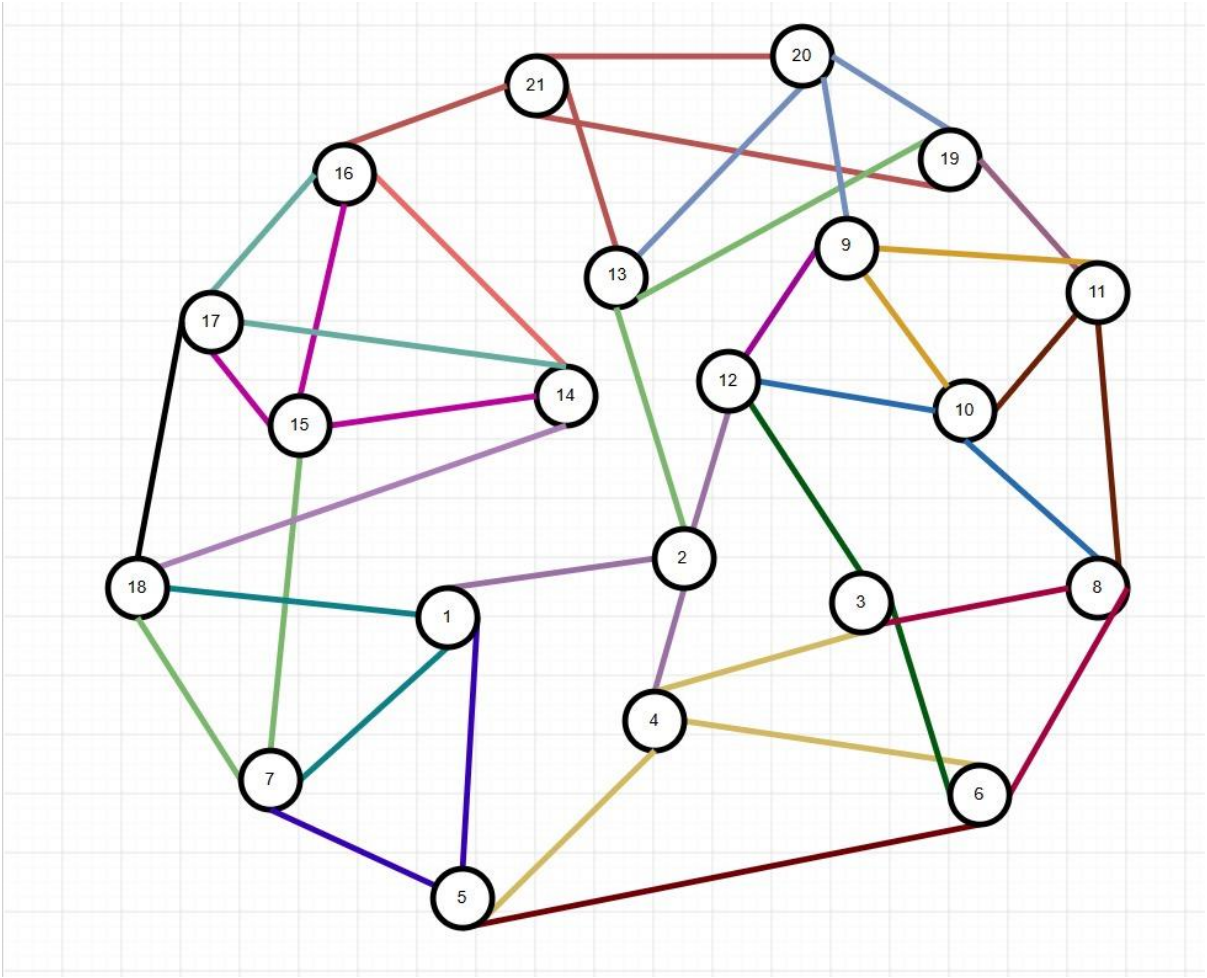
12: [9, 2, 3, 10],

```
    13: [21, 20, 2, 19],
    14: [18, 15, 16, 17],
    15: [14, 16, 17, 7],
    16: [14, 15, 17, 21],
    17: [14, 15, 16, 18],
    18: [1, 7, 14, 17],
    19: [11, 13, 20, 21],
    20: [9, 19, 13, 21],
    21: [13, 16, 20, 19]
}

# Exibir as conexões e o grau de cada vértice
print("--- CONEXÕES E GRAUS DE CADA VÉRTICE ---")
for vertice, conexoes in grafo.items():
    grau = len(conexoes)
    print(f"Vértice {vertice} -> Conexões: {conexoes} | Grau: {grau}")

# Exibir estatísticas gerais do grafo
print("\n--- ESTATÍSTICAS DO GRAFO ---")
print("Total de vértices:", len(grafo))

# Soma de todos os graus dividido por 2 (pois cada aresta liga dois vértices)
total_graus = sum(len(conexoes) for conexoes in grafo.values())
total_arestas = total_graus // 2
print("Total de arestas:", total_arestas)
```


20
21
19
16
9
13
11
17
12
14
10
15
2
18
3
8
1
4
7
6
5